

Joydeep Banik

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EDUCATION

- RCC Institute of Information Technology** Kolkata, India
Bachelor of Technology - Electronics and Communication Engineering; SGPA: 7.32 August 2023 - August 2027
Courses: Operating Systems, Data Structures and Algorithms, Design and Analysis Of Algorithms, Digital Signal Processing, Computer Network, Information Theory and Coding

SKILLS SUMMARY

- Languages:** Python, C++, Java, JavaScript, TypeScript, HTML5, CSS3, Bash, MATLAB
- Backend and Database:** Django, Flask, NodeJS, Relational Databases(MySQL, SQLite), Neo4j
- Data & ML Libraries:** TensorFlow, PyTorch, OpenCV, NumPy, Rasterio, Matplotlib
- Cloud & Developer Tools:** Kubernetes, Docker, Apache Kafka, Git, GCP

EXPERIENCE

- Jadavpur University, Dept. of ETCE** On-Site
Research Intern May 2026 - Present
 - Real-time CNN dehazing pipeline:** Optimized a real-time CNN dehazing pipeline in Python, reducing frame latency through efficient pre-processing and model acceleration.
 - RAG Framework for Video Dehazing:** Architecting an end-to-end Retrieval-Augmented Generation framework that integrates latent-space feature retrieval pipelines to improve frame-level clarity and preserve temporal consistency across video sequences.
- Haldia Petrochemicals Ltd.** On-Site
Information Systems Intern December, 2025 - January, 2026
 - Role-Based Store Management Engine:** Built an API-driven inventory system in Python using object-oriented design to automate tracking across procurement, warehouse, and logistics departments.
 - Transactional Reliability:** Implemented unit-tested database transactions that eliminated manual inventory errors and enforced production-grade code quality.
 - Network Infrastructure:** Audited a 3-layer enterprise network architecture and leveraged PRTG telemetry to accelerate infrastructure incident triage and boost system uptime.

PROJECTS

- KubeSense** Live
Terminal-based Kubernetes Cluster Monitor GitHub
 - Distributed Monitoring:** Engineered a terminal-based (TUI) application for real-time distributed Kubernetes cluster monitoring, featuring live container log tailing and multi-node SSH metric fallback routing for zero-GUI diagnostics.
 - Anomaly Detection:** Implemented passive AI-driven log parsing for dynamic anomaly detection, enabling administrators to triage infrastructure incidents and evaluate pod telemetry without a graphical interface.
- AgriSense AI** Smart India Hackathon 2025
Automated Satellite Crop-Stress Analytics GitHub
 - Satellite Pipeline:** Developed an automated satellite analytics pipeline using Google Earth Engine and Python to process Landsat 9 telemetry across multi-hectare farmland.
 - Stress Detection:** Applied CNN and K-Means algorithms to detect vegetation stress, exposing results through an interactive web dashboard for continuous field-level analysis.
- Sovereign Bharat Intelligence Graph** India Innovates 2026
Global Ontology Engine / Graph-RAG Pipeline GitHub
 - Data Ingestion:** Engineered a distributed, fault-tolerant data ingestion pipeline using Apache Kafka to unify structured, unstructured, and real-time feeds into a Neo4j-backed intelligence graph.
 - Querying Layer:** Built a Graph-RAG querying layer with Gemma 3 LLM, linking cross-domain data—geopolitics, economics, climate, defense—for early risk detection and scenario simulation.

PUBLICATIONS

- A Review of Dehazing Benchmark Datasets:** Accepted for presentation, 15th International Conference on Image Processing Theory, Tools and Applications (IPTA), December, 2026

HONORS AND AWARDS

- Selected as a national finalist at India Innovates 2026 for 'Sovereign Bharat Intelligence Graph'.
- Recognized as a national semifinalist at Smart India Hackathon 2025 for 'AgriSense AI'.

VOLUNTEER EXPERIENCE

- Amplify RCCIIT** Kolkata, India
Event Coordinator November 2025
 - Built full-stack Capture The Flag infrastructure including scoring logic, challenge deployment pipelines, and live monitoring dashboards.
- RCC Talkies RCCIIT** Kolkata, India
Research Wing Member 2024 - 2025
 - Contributing feature articles on emerging technology trends for the institution's technical magazine.